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EDUCATION

Ph.D. – Computer Science, University of Tennessee, Knoxville, Tennessee, U.S.A., **1998**

Dissertation: *Stochastic models for performance analyses of iterative algorithms in distributed environments*

Advisors: Prof. Jack J. Dongarra and Prof. Michael G. Thomason

M.S. – Diplôme d'Études Approfondies (DEA) in Parallel Architectures and Applied Mathematics, National Polytechnic Institute of Toulouse, France, **1994**.

Thesis: *A version of PVM for shared- and virtually shared-memory supercomputers*

Advisor: Prof. Michel Daydé

B.S. – Diplôme d'Ingénieur in Computer Science and Applied Mathematics, Ecole Nationale Supérieure d'Electronique, d'Electrotechnique, d'Informatique et d'Hydraulique de Toulouse (ENSEEIHT), Toulouse, France, **1993**.

RESEARCH INTERESTS

High performance computing, parallel and distributed computing, parallel application scheduling, simulation of distributed applications and platforms.

PROFESSIONAL EXPERIENCE

07/15 - present: Professor, Information and Computer Sciences Department, University of Hawai'i at Mānoa.

04/12 - 03/18: Visiting Professor, National Institute of Informatics, Tokyo, Japan.

07/07 - 06/15: Associate Professor, Information and Computer Sciences Department, University of Hawai'i at Mānoa.

08/05 - 06/07: Assistant Professor, Information and Computer Sciences Department, University of Hawai'i at Mānoa.

07/04 - 07/05: Associate Research Scientist, San Diego Supercomputer Center, University of California, San Diego.

07/02 - 07/05: Adjunct Assistant Professor, Department of Computer Science and Engineering, University of California, San Diego.

07/01 - 06/04: Assistant Research Scientist, San Diego Supercomputer Center, University of California San Diego.

07/01 - 07/05: Director of the Grid Research And Innovation Laboratory (GRAIL), University of

California San Diego.

12/98 - 06/01: Assistant Project Scientist, Department of Computer Science and Engineering, University of California, San Diego.

03/98 - 11/98: Postdoctoral Research Associate, Computer Science Department, University of Tennessee, Knoxville.

01/95 - 02/98: Graduate Research Assistant, Computer Science Department, University of Tennessee, Knoxville.

11/93 - 11/94: Computer Science Advisor, French Ministry of Defense (DGA), Paris, France.

09/92 - 10/93: Graduate Research Assistant, Computer Science Research Institute of Toulouse (IRIT), Toulouse, France.

PUBLICATIONS

Journal Articles

63. *Lowering Entry Barriers to Developing Custom Simulators of Distributed Applications and Platforms with SimGrid*, H. Casanova, A. Giersch, A. Legrand, M. Quinson, F. Suter, *Parallel Computing*, 123, 103–125, **2025**.
62. *An Exploration of Online-simulation-driven Portfolio Scheduling in Workflow Management Systems*, J. McDonald, J. Dobbs, Y. C. Wong, R. Ferreira da Silva, H. Casanova, *Future Generation Computer Systems (FGCS)*, 161, 345–360, **2024**.
61. *Automated Generation of Scientific Workflow Generators with WfChef*, T. Coleman, H. Casanova, R. Ferreira da Silva, *Future Generation Computer Systems (FGCS)*, 147, 16–29, **2023**.
60. *Beyond Binary Search: Parallel In-place Construction of Implicit Search Tree Layouts*, K. Berney, H. Casanova, B. Karsin, N. Sitchinava, *IEEE Transactions on Computers (TC)*, 71(5), 1104–1116, **2022**.
59. *WfCommons: A Framework for Enabling Scientific Workflow Research and Development*, T. Coleman, H. Casanova, L. Pottier, M. Kaushik, E. Deelman, R. Ferreira da Silva, *Future Generation Computer Systems (FGCS)*, 128, 16–27, **2021**.
58. *Teaching Parallel and Distributed Computing Concepts in Simulation with WRENCH*, H. Casanova, R. Tanaka, W. Koch, R. Ferreira da Silva, *Journal of Parallel and Distributed Computing (JPDC)*, 156, 53–63, **2021**.
56. *Characterizing, Modeling, and Accurately Simulating Power and Energy Consumption of I/O-intensive Scientific Workflows*, R. Ferreira da Silva, H. Casanova, A.-C. Orgerie, R. Tanaka, E. Deelman F. Suter, *Journal of Computational Science (JOCS)*, 44, **2020**.
55. *Developing Accurate and Scalable Simulators of Production Workflow Management Systems with WRENCH*, H. Casanova, R. Ferreira da Silva, R. Tanaka, S. Pandey, G. Jethwani, W. Koch, S. Albrecht, J. Oeth, F. Suter, *Future Generation Computer Systems (FGCS)*, 112, 162–175, **2020**.
54. *MILP Formulations for Spatio-Temporal Thermal-Aware Scheduling in Cloud and HPC Datacenters*, J.-M. Pierson, P. Stolf, H. Sun, H. Casanova, in *Cluster Computing*, **2019**.
53. *An Efficient Algorithm for the 1D Total Visibility-Index Problem*, P. Afshani, M. de Berg, H. Casanova, B. Karsin, C. Lambrechts, N. Sitchinava, C. Tsirogiannis, in *Journal of Experimental Algorithmics*, 23(2.3), 1–23, **2018**.

52. *Checkpointing Workflows for Fail-Stop Errors*, L. Han, L.-C. Canon, H. Casanova, Y. Robert, F. Vivien, in *IEEE Transactions on Computers (TC)*, 67(8), 1105–1120, **2018**.

51. *Computing the expected makespan of task graphs in the presence of silent errors*, J. Herrmann, Y. Robert, H. Casanova, in *Parallel Computing*, 75, 41–60, **2018**.

50. *Checkpointing Strategies for Scheduling Computational Workflows*, G. Aupy, A. Benoit, H. Casanova, Y. Robert, in International *Journal of Networking and Computing*, 6(1), 2–26, **2016**.

49. *Distance Threshold Similarity Searches: Efficient Trajectory Indexing on the GPU*, M. Gowanlock, H. Casanova, in *IEEE Transactions on Parallel and Distributed Systems*, 27(9), 2533–2545, **2016**.

48. *On the Impact of Process Replication on Executions of Large-Scale Parallel Applications with Coordinated Checkpointing*, H. Casanova, Y. Robert, F. Vivien, D. Zaidouni, in *Future Generation Computer Systems*, 51, 7–19, **2015**.

47. *Toward More Scalable Off-Line Simulations of MPI Applications*, H. Casanova, A. Gupta, F. Suter, in *Parallel Processing Letters*, 25(3), **2015**.

46. *Simulation of MPI Applications with Time-Independent Traces*, F. Deprez, S. Markomanolis, F. Suter, H. Casanova, in *Concurrency and Computation: Practice and Experience*, 27(5), 1145–1168, **2015**.

45. *Selecting Linear Algebra Kernel Composition Using Response Time Prediction*, A. Hurault, K. Baek, H. Casanova, in *Software: Practice and Experience*, 45(12), 1659–1676, **2015**.

44. *In-Memory Distance Threshold Similarity Searches on Moving Object Trajectories*, M. Gowanlock, H. Casanova, in *International Journal On Advances in Software*, 7 (3-4), 617–631 **2014**.

43. *Swap-and-randomize: A Method for Building Low-latency HPC Interconnects*, I. Fujiwara, M. Koibuchi, H. Matsutani, H. Casanova, in *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 26(7), 2051–2060, **2014**.

42. *Versatile, Scalable, and Accurate Simulation of Distributed Applications and Platforms*, H. Casanova, A. Giersch, A. Legrand, M. Quinson, F. Suter, in *Journal of Parallel and Distributed Computing*, 75(10), 2899–2917, **2014**.

41. *Mapping Applications on Volatile Resources*, H. Casanova, F. Dufossé, Y. Robert, F. Vivien, in *International Journal of High Performance Computing Applications (IJHPCA)*, 29(1), 73–91, **2014**.

40. *Using group replication for resilience on exascale systems*, M. Bougeret, H. Casanova, Y. Robert, F. Vivien, D. Zaidouni, in *International journal of High Performance Computing Applications (IJHPCA)*, 28(2), 210–224, **2014**.

39. *On the Validity of Flow-level TCP Network Models for Grid and Cloud Simulations*, P. Velho, L. Mello Schnorr, H. Casanova, A. Legrand, in *ACM Transactions on Modeling and Computer Simulation (TOMACS)*, 23(4), **2013**.

38. *Energy-Aware Service Allocation*, D. Borgetto, H. Casanova, G. Da Costa, J.M. Pierson, in *Future Generation Computer Systems (FGCS)*, 28(5), 769–779, **2012**.

37. *Dynamic Fractional Resource Scheduling vs. Batch Scheduling*, M. Stillwell, F. Vivien, H. Casanova, in *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 23(3), 521–529, **2012**.

36. *Preventive Migration vs. Preventive Checkpointing for Extreme Scale Supercomputers*, F. Cap-

pello, H. Casanova, Y. Robert, in *Parallel Processing Letters*, 21(2), 111–132, **2011**.

35. *Resource Allocation for Multiple Concurrent In-network Stream-processing Applications*, A. Benoit, H. Casanova, V. Rehn-Sonigo, Y. Robert, in *Parallel Computing (ParCo)*, 30, 331–348, **2011**.
34. *On Cluster Resource Allocation for Multiple Parallel Task Graphs*, H. Casanova, F. Desprez, F. Suter, in the *Journal of Parallel and Distributed Computing (JPDC)*, 70(12), 1193–1203, **2010**.
33. *Resource Allocation Algorithms for Virtualized Service Hosting Platforms*, M. Stillwell, D. Schanzenbach, F. Vivien, H. Casanova, in the *Journal of Parallel and Distributed Computing (JPDC)*, 70(9), 962–974, **2010**.
32. *Resource Allocation Strategies for Constructive In-Network Stream Processing*, A. Benoit, V. Rhen-Sonigo, Y. Robert, H. Casanova, in the *International Journal of Foundations of Computer Science (IJFCS)*, 22(3), 621-638, **2010**.
31. *Characterizing Fault Tolerance in Genetic Programming*, D. Lombraña González, F. Fernández de Vega, H. Casanova, in the *Journal of Future Generation Computer Systems (FGCS)*, 26, 847–856, **2010**.
30. *Scheduling Mixed-Parallel Applications with Advance Reservations*, K. Aida, H. Casanova, in the *Cluster Computing Journal*, 12(2), 205–220, **2009**.
29. *Scheduling Parallel Task Graphs on (Almost) Homogeneous Multi-cluster Platforms*, P.-F. Dutot, T. N'takpé, F. Suter, H. Casanova, in *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 20(7), 940–952, **2009**.
28. *Scheduling Task Parallel Applications for Rapid Application Turnaround on Enterprise Desktop Grids*, D. Kondo, A. Chien, H. Casanova, in the *Journal of Grid Computing*, 5(4), 379–405, **2007**.
27. *Benefits and Drawbacks of Redundant Batch Requests*, H. Casanova, in the *Journal of Grid Computing*, 5(2), 235-250, **2007**
26. *Characterizing Resource Availability in Enterprise Desktop Grids*, D. Kondo, G. Fedak, F. Cappello, H. Casanova, in the *Journal of Future Generation Computer Systems*, 23(7), 888–903, **2007**.
25. *Steady-State Scheduling of Multiple Divisible Load Applications on Wide-Area Distributed Computing Platforms*, L. Marchal, Y. Yang, H. Casanova, Y. Robert, in the *International Journal of High Performance Computing Applications* (IJHPCA), 20(8), 365–381, **2006**.
24. *Interference-Aware Scheduling*, B. Kreaseck, L. Carter, H. Casanova, J. Ferrante, S. Nandy, in the *International Journal of High Performance Computing Applications* (IJHPCA), 20(2), 45–59, **2006**.
23. *New Grid Scheduling and Rescheduling Methods in the GrADS Project*, F. Berman, H. Casanova, A. Chien, K. Cooper, H. Dail, A. Dasgupta, W. Deng, J. Dongarra, L. Johnsson, K. Kennedy, C. Koelbel, B. Liu, X. Liu, A. Mandal, G. Marin, M. Mazin, J. Mellor-Crummey, C. Mendes, A. Olugbile, M. Patel, D. Reed, Z. Shi, O. Sievert, H. Xia, A. Yarkhan, in the *International Journal of Parallel Programming*, 33(2), 209–229, **2005**.
22. *Network Modeling Issues for Grid Application Scheduling*, H. Casanova, in the *International Journal of Foundations of Computer Science (IJFCS)*, 16(2), 145–162, **2005**.
21. *Multi-Round algorithms for Scheduling Divisible Workloads*, Y. Yang, K. van der Raadt, H.

Casanova, *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 16(11), 1092–1102, **2005**.

20. *Scheduling Divisible Loads on Star and Tree Networks: Results and Open Problems*, O. Beaumont, H. Casanova, A. Legrand, Y. Robert, Y. Yang, *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 16(3), 207–218, **2005**.

19. *The Encyclopedia of Life Project: Grid Software and Deployment*, W. Li, R. Byrnes, J. Hayes, V. Reyes, A. Birnbaum, A. Shabab, C. Mosley, D. Pekurowsky, G. Quinn, I. Shindyalov, H. Casanova, L. Ang, F. Berman, M. Miller, P. Bourne, in the *Journal of New Generation Computing on Grid Systems for Life Sciences*, 22(2), 127–136, **2004**.

18. *A Simple MPI Process Swapping Architecture for Iterative Applications*, O. Sievert, H. Casanova, *The International Journal of High Performance Computing Applications* (IJHPCA), 18(3), 341–352, **2004**.

17. *The Virtual Instrument: Support for Grid-enabled Scientific Simulations*, H. Casanova, et al., *The International Journal of High Performance Computing Applications* (IJHPCA), 18(1), 3–17, **2004**.

16. *Resource Allocation Strategies for Guided Parameter Space Searches*, M. Faerman, A. Birnbaum, F. Berman, H. Casanova, *The International Journal of High Performance Computing Applications* (IJHPCA), 17(4), 383–402, **2003**.

15. *Adaptive Computing on the Grid Using AppLeS*, F. Berman, R. Wolski, H. Casanova, W. Cirne, H. Dail, M. Faerman, S. Figueira, J. Hayes, G. Obertelli, J. Schopf, G. Shao, S. Smallen, N. Spring, A. Su, D. Zagorodnov, *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 14(4), 369–382, **2003**.

14. *A Decoupled Scheduling Approach for Grid Application Development Environments*, H. Dail, F. Berman, H. Casanova, *Journal of Parallel and Distributed Computing* (JPDC), Special issue on Grid Computing, 63(5), 505–524, **2003**.

13. *Using TOP-C and AMPIC to Port Large Parallel Applications to the Computational Grid*, G. Cooperman, H. Casanova, J. Hayes, T. Witzel, *The International Journal on Future Generation Computer Systems* (FGCS), 19(4), 587–596, **2003**.

12. *Applying Scheduling and Tuning to On-line Parallel Tomography*, S. Smallen, H. Casanova, F. Berman, *Scientific Programming*, 10(4), **2003**.

11. *Innovation of the NetSolve Grid Computing System*, D. Arnold, H. Casanova, J. Dongarra, *Concurrency and Computation: Practice and Experience* (CCPE), 14(13-15), 1457–1479, **2002**.

10. *Middleware for the Use of Storage in Communication*, M. Beck, D. Arnold, A. Bassi, F. Berman, H. Casanova, J. Dongarra, T. Moore, G. Obertelli, J. Plank, M. Swany, S. Vadhiyar, R. Wolski, *Parallel Computing*, 28(12), 1773–1787, **2002**.

9. *Distributing MCell Simulations on the Grid*, H. Casanova, T. Bartol, J. Stiles, F. Berman, *The International Journal of High Performance Computing Applications* (IJHPCA), Vol. 14(3), 243–257, Fall **2001**.

8. *The AppLeS Parameter Sweep Template: User-Level Middleware for the Grid*, H. Casanova, G. Obertelli, F. Berman, R. Wolski, *Scientific Programming*, Vol. 8(3), 111–126, **2001**.

7. *Adaptive Scheduling for Task Farming with Grid Middleware*, H. Casanova, M. Kim, J. Plank, J. Dongarra, *The International Journal of Supercomputer Applications and High Performance Computing* (IJHPCA), Vol. 13, 231–240, **1999**.

6. *Logistical Quality of Service in NetSolve*, M. Beck, H. Casanova, J. Dongarra, T. Moore, J. Plank, F. Berman, R. Wolski, *Computers and Communications*, Vol. 22, Issue 11, 1034–1044, **1999**.
5. *Deploying Fault tolerance and Task Migration with NetSolve*, J. Plank, H. Casanova, M. Beck, J. Dongarra, *The International Journal on Future Generation Computer Systems* (FGCS), Vol. 15, 745–755, **1999**.
4. *Stochastic Performance Prediction for Iterative Algorithms in Distributed Environments*, H. Casanova, M. Thomason, J. Dongarra, *Journal of Parallel and Distributed Computing* (JPDC), Vol. 58(1), 68–91, **1999**.
3. *Using agent-based software for scientific computing in the NetSolve system*, H. Casanova, J. Dongarra, *Parallel Computing*, Vol. 24, 1777–1790, **1998**.
2. *Java Access to Numerical Libraries*, H. Casanova, D. Doolin, J. Dongarra, *Concurrency: Practice and Experience* (CPE), Vol. 9(11), 1279–1291, **1997**.
1. *NetSolve: A Network-Enabled Server for Solving Computational Science Problems* , H. Casanova, J. Dongarra, *The International Journal of Supercomputer Applications and High Performance Computing* (IJHPCA), Vol. 11(3), 212–223, Fall **1997**.

Refereed Conference and Workshop Proceedings

115. *Determining Levels of Detail for Simulators of Parallel and Distributed Computing Systems via Automated Calibration*, J. McDonald, Y.-C. Wong, K. Mehta, F. Suter, R. Ferreira da Silva, L. Pottier, E. Deelman, H. Casanova, in Proc. of the 16th International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems (PMBS), Nov **2025**.
114. *Simulation of Large-Scale HPC Storage Systems: Challenges and Techniques*, J. Monniot, F. Tessier, H. Casanova, G. Antoniu, in Proc. of the 31st IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC), Distinguished paper award finalist, Dec **2024**.
113. *Automated Calibration of Parallel and Distributed Computing Simulators: A Case Study*, J. McDonald, M. Horzela, F. Suter, H. Casanova, in Proc. of the 25th IEEE International Workshop on Parallel and Distributed Scientific and Engineering Computing (PDSEC), May **2024**.
112. *Modelling Distributed Heterogeneous Computing Infrastructures for HEP Applications*, M. Horzela, H. Casanova, M. Giffels, A. Gottmann, R. Hofsaess, G. Quast, S. Rossi Tisbeni, A. Streit, F. Suter, in Proc. of the 26th International Conference on Computing in High Energy and Nuclear Physics (CHEP), May **2023**.
111. *WfCommons: Data Collection and Runtime Experiments using Multiple Workflow Systems*, H. Casanova, K. Berney, S. Chastel, R. Ferreira da Silva, in Proc. of the 1st IEEE International Workshop on Workflows in Distributed Environments (WiDE), Jun **2023**
110. *WfBench: Automated Generation of Scientific Workflow Benchmarks*, T. Coleman, H. Casanova, K. Maheswari, L. Pottier, S. Wilkinson, J. Wozniak, F. Suter, M. Shankar, R. Ferreira da Silva, in Proc. of the 13th International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer System (PBMS), Nov **2022**
109. *On the Feasibility of Simulation-driven Portfolio Scheduling for Cyberinfrastructure Runtime*

Systems, H. Casanova, Y. C. Wong, L. Pottier, R. R. Ferreira da Silva, in Proc. of the 25th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), June **2022**

108. *Peachy Parallel Assignments (EduPar 2022)*, H. M. Bücker, H. Casanova, R. Ferreira da Silva, A. Lasserre, D. Luyen, R. Namyst, J. Schoder, P.-A. Wacrenier, D. P. Bunde, Proc. of the Workshop on Parallel and Distributed Computing Education (EduPar), Jun **2022**.

107. *Peachy Parallel Assignments (EduHPC 2021)*, H. Casanova. R. Ferreira da Silva, A. Gonzalez-Escribano, H. Li, Y. Torres, and D. Bunde, in proc. of the Workshop on Education for High-Performance Computing (EduHPC), Nov **2021**.

106. *Modeling the Linux page cache for accurate simulation of data-intensive applications*, H.-D. Do, V. Hayot-Sasson, R. Ferreira da Silva, C. Steele, H. Casanova, T. Glatard, in Proc. of IEEE Cluster, Sep. **2021**

105. *WfChef: Automated Generation of Accurate Scientific Workflow Generators* , T. Coleman, H. Casanova, R. Ferreira da Silva, in Proc. of the 17th IEEE International eScience Conference (eScience), Sep. **2021**

104. *GLUME: A Strategy for Reducing Workflow Execution Times on Batch-Scheduled Platforms*, E. Hataishi, P.-F. Dutot, R. Ferreira da Silva, H. Casanova, in Proc. of the 24th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), May **2021**

103. *Evaluating energy-aware scheduling algorithms for I/O-intensive scientific workflows*, T. Coleman, H. Casanova, T. Gwartney, R. Ferreira da Silva, in Proc. of the International Conference on Computational Science (ICCS), June **2021**

102. *Peachy Parallel Assignments (EduHPC 2020)*, H. Casanova, R. Ferreira Da Silva, A. Gonzalez-Escribano, W. Koch, Y. Torres, and D.P. Bunde, in Proc. of the Workshop on Education for High-Performance Computing (EduHPC), Nov. **2020**.

101. *WorkflowHub: Community Framework forEnabling Scientific Workflow Research and Development*, R. Ferreira Da Silva, Pottier, L., Coleman, T., Deelman, E., H. Casanova, in Proc. of the 15th Workshop on Workflows in Support of Large-Scale Science (WORKS), Nov. **2020**.

100. *Modeling the Performance of Scientific Workflow Executions on HPC Platforms with Burst Buffers*, L. Pottier, R. Ferreira Da Silva, H. Casanova, E. Deelman, in Proc. of the 22nd IEEE Cluster conference, Kobe, Japan, Sep. **2020**.

99. *Teaching Parallel and Distributed Computing Concepts in Simulation with WRENCH*, R. Tanaka, R. Ferreira da Silva, H. Casanova, in Proc. of the Workshop on Education for High-Performance Computing (EduHPC), Denver, Colorado, Nov. **2019**.

98. *Bridging Concepts and Practice in eScience via Simulation-driven Engineering*, R. Ferreira da Silva, H. Casanova, R. Tanaka, F. Suter, in Proc. of the Bridging from Concepts to Data and Computation for eScience Workwhop (BC2DC), Sep. **2019**.

97. *Accurately Simulating Energy Consumption of I/O-intensive Scientific Workflows*, R. Ferreira da Silva, A.-C. Orgerie, H. Casanova, R. Tanaka, E. Deelman, F. Suter, in Proc. of the International Conference on Computational Science (ICCS), Faro, Portugal, June **2019**.

96. *Sparse 3-D NoCs with Inductive Coupling*, M. Koibuchi, L. Leong, T. Totoki, H. Matsutani, H. Amana, H. Casanova, in Proc. of the Design Automation Conference (DAC), Las Vegas, Texas, June **2019**.

95. *WRENCH: Workflow Management System Simulation Workbench*, H. Casanova, S. Pandey, J. Oeth, R. Tanaka, F. Suter, R. Ferreira da Silva, in Proc. of the Workshop on Workflows in

Support of Large-Scale Science (WORKS), Dallas, Texas, Nov. **2018**.

94. *SMPI Courseware: Teaching Distributed-Memory Computing with MPI in Simulation*, H. Casanova, M. Quinson, A. Legrand, F. Suter, in Proc. of the Workshop on Education for High-Performance Computing (EduHPC), **best paper award**, Dallas, Texas, Nov. **2018**.
93. *Analysis-driven Engineering of Comparison-based Sorting Algorithms on GPUs*, B. Karsin, V. Weichert, H. Casanova, J. Iacono, N. Sitchinava, in Proc. of the 32nd ACM International Conference on Supercomputing (ICS), Beijing, China, June **2018**.
92. *Beyond binary search: parallel in-place construction of implicit search trees*, K. Berney, H. Casanova, A. Higuchi, B. Karsin, N. Sitchinava, in Proc. of the 32nd IEEE International Parallel & Distributed Processing Symposium (IPDPS), Vancouver, Canada, May **2018**.
91. *Checkpointing Workflows for Fail-Stop Errors*, L. Han, L.-C. Canon, H. Casanova, Y. Robert, F. Vivien, in Proc. of the 19th IEEE Cluster conference, Honolulu, Hawai'i, Sep. **2017**.
90. *A Case for Uni-Directional Network Topologies in Large-Scale Clusters*, M. Koibuchi, T. Totoki, H. Matsutani, H. Amano, F. Chaix, I. Fujiwara, H. Casanova, in Proc. of the 19th IEEE Cluster conference, Honolulu, Hawai'i, Sep. **2017**.
89. *High-Bandwidth Low-Latency Approximate Interconnection Networks*, D. Fujiki, K. Ishii, I. Fujiwara. H. Matsutani, H. Amano, H. Casanova, M. Koibuchi, in Proc. of the 23rd IEEE Symposium on High Performance Computer Architecture (HPCA), Austin, Texas, Feb. **2017**.
88. *An Efficient Algorithm for the 1D Total Visibility-Index Problem*, P. Afshani, M. de Berg, H. Casanova, B. Karsin, C. Lambrechts, N. Sitchinava, C. Tsirogiannis, in Proc. of Algorithm Engineering & Experiments (ALENEX), Barcelona, Spain, Jan. **2017**.
87. *Towards Ideal Hop Counts in Interconnection Networks with Arbitrary Size*, M. Koibuchi, I. Fujiwara, F. Chaix, H. Casanova, in Proc. of the 4th International Symposium on Computing and Networking (CANDAR), Hiroshima, Japan, Nov. **2016**.
86. *Computing the expected makespan of task graphs in the presence of silent errors*, H. Casanova, J. Herrmann, Y. Robert, in Proc. of the 9th International Workshop on Programming Models and Systems Software for High-End Computing, Philadelphia, Pennsylvania, Aug. **2016**.
85. *Efficient Batched Predecessor Search in Shared Memory on GPUs*, B. Karsin, H. Casanova, N. Sitchinava, in Proc. of the 22nd IEEE International Conference on High Performance Computing (HiPC), Bengaluru, India, Dec. **2016**.
84. *Scheduling computational workflows on failure-prone platforms*, G. Aupy, A. Benoit, H. Casanova, Y. Robert, in Proc. of the 17th Workshop on Advances on Parallel and Distributed Processing Symposium (APDCM), Hyderabad, India, May **2015**.
83. *Indexing of Spatiotemporal Trajectories for Efficient Distance Threshold Similarity Searches on the GPU*, M. Gowanlock, H. Casanova, in Proc. of the 29th IEEE International Parallel & Distributed Processing Symposium (IPDPS), Hyderabad, India, May **2015**.
82. *Augmenting Low-latency HPC Network with Free-space Optical Links*, I. Fujiwara, M. Koibuchi, T. Ozaki, H. Matsutani, H. Casanova, in Proc. of the 21st IEEE International Symposium on High Performance Computer Architecture (HPCA), Bay Area, California, Feb. **2015**.
81. *Distance Threshold Similarity Searches on Spatiotemporal Trajectories using GPGPU*, M. Gowanlock, H. Casanova, in Proc. of the 21st IEEE Conference on High Performance Computing (HiPC), Goa, India, Dec. **2014**.
80. *In-Memory Distance Threshold Queries on Moving Object Trajectories*, M. Gowanlock, H.

Casanova, in Proc. of the 6th International Conference on Advances in Databases, Knowledge, and Data Applications (DBKDA), Chamonix, France, Apr. **2014**.

79. *Cost-Optimal Execution of Boolean Query Trees with Shared Streams*, H. Casanova, L. Lim, Y. Robert, F. Vivien, D. Zaidouni, in Proc. of 28th IEEE International Parallel and Distributed Processing Symposium (IPDPS), Phoeniz, AZ, May **2014**.

78. *Skywalk: a Topology for HPC Networks with Low-delay Switches*, I. Fujiwara, M. Koibuchi, H. Matsutani, H. Casanova, in Proc. of 28th IEEE International Parallel and Distributed Processing Symposium (IPDPS), Phoeniz, AZ, May **2014**.

77. *Scheduling Tightly-Coupled Applications on Heterogeneous Desktop Grids*, H. Casanova, F. Dufossé, Y. Robert, F. Vivien, in Proc. of the 21st International Heterogeneity in Computing Workshop (HCW), Boston, MA, May **2013**.

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Misc. Publications and White Papers

5. *Generating Grid Resource Requirement Specifications*, R. Huang, H. Casanova, A. Chien, 2-page poster in Proc. of the 16th IEEE International Symposium on High Performance Distributed Computing (HPDC-16), Monterey, California, Jun. **2007**
4. *Robust Resource Allocation for Large-scale Distributed Shared Resource Environments*, Y.-S. Kee, K. Yocum, A. Chien, and H. Casanova, 2-page poster in Proc. of the 15th IEEE International Symposium on High Performance Distributed Computing (HPDC-15), Paris,

France, Jun. **2006**

3. *The* SimGrid *Project: Simulation and Deployment of Distributed Applications*, A. Legrand, M. Quinson, H. Casanova, K. Fujiwara, 2-page poster in Proc. of the 15th IEEE International Symposium on High Performance Distributed Computing (HPDC-15), Paris, France, Jun. **2006**
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INVITED PRESENTATIONS, TUTORIALS, and COLLOQUIA

- *Insights and Challenges for the Simulation of Parallel and Distributed Computing Systems: A SimGrid Perspective*, **Keynote Address**, IEEE/SBC 36th International Symposium on Computer Architecture and High Performance Computing, Hilo, Hawai'i, Nov. 2024.
- *On the use of simulation for scheduling (research)*, 15th Scheduling for Large Scale Systems Workshop, Fréjus, France, June 2022.
- *Checkpointing Workflows for Fail-Stop Failures*, Information Science Institute, Los Angeles, California, Nov. 2018.
- *Scheduling and Simulating Parallel and Distributed Applications*, 6-session seminar series, National Institute for Informatics, Tokyo, Japan, Oct. 2013.
- *Cost-Optimal Execution of Boolean Query Trees with Shared Streams*, INRIA Workshop on Scheduling for Exascale Systems, Dagstuhl Schloss, Germany, Sep. 2013.
- *A Case for Random Topologies in HPC Interconnects*, INRIA Workshop on Scheduling, Pittsburgh, Pennsylvania, Jun. 2012.
- *Challenges and Hopes for Scheduling Parallel Applications on Desktop Grids*, **Keynote Address**, Workshop on Desktop Grids and Volunteer Computing Systems (PCGrid'11), Anchorage, Alaska, May 2011.
- *Flexible Cluster Sharing*, Institut de Recherche en Informatique de Toulouse, Toulouse, France, Nov. 2008.
- *Parallel resource sharing: Shortcomings, Cheats, and Perspectives*, Sept. of Computer Science, University of Bologna, Bologna, Italy, Jun. 2008.
- *Scheduling Mixed-Parallel Applications with Advance Reservations*, INRIA Workshop on scheduling for large scale distributed platforms, Aussois, France, May 2008.
- *Simulation for Large-Scale Distributed Computing Research*, tutorial at the 8th IEEE International Symposium on Cluster Computing and the Grid (CCGrid'08), Lyon, France, May 2008.
- *Grid Computing: Technology and Applications*, invited plenary presentations at Telecom'05, Hawai'i TeleCommunication Association, Honolulu, Hawai'i, Oct. 2005.
- *Scheduling and Deploying Applications on Large-Scale Platforms*, invited lecture at the Departamento de Informática, Centro Universitario de Mérida, Mérida, Spain, Jun. 2005.
- *Simulation for Grid Computing*, **Keynote Address**, 2nd International Grid Performance Workshop, Edinburgh, United Kingdom, Jun. 2005.

- *Grid Computing: Principles, Software, Applications, and Tools*, **Keynote Address**, 2nd Workshop on New Frontiers of Bioinformatics in Latin America – Gridding Biology, Merida, Venezuela, Nov. 2004.
- *Grid platform modeling issues for scheduling research* , Workshop on scheduling for large scale distributed platforms, Aussois, France, Aug. 2004.
- *Grid Technology, Application, and Research*, presentation to the IEEE Society, University of California, La Jolla, California, May 2004.
- *Deploying Applications on the Grid: Engineering and Research*, University of San Diego, San Diego, California, Sep. 2003.
- *Divisible Workload Scheduling*, Paris XI University, Orsay, France, Jun. 2003.
- *UMR and RUMR: Algorithms to Schedule Divisible Workloads*, the Tokyo Institute of Technology (TITECH), Tokyo, Japan, May 2003.
- *MCell on the Grid: applications, software, research, and impact*, Plenary presentation at the National Parternship for Advanced Computing Infrastructure All-Hands Meeting (NPACI-AHM), La Jolla, California, Mar. 2003.
- *From Parameter Sweep Applications to Divisible Workloads*, Workshop on Distributed Computing, Metacomputing, and Resource Globalization, Aussois, France, Nov. 2002.
- *An Introduction to Grid Computing*, the Salk Institute, La Jolla, California, Nov. 2002.
- *Processor Swapping for Enterprise Computing*, Workshop on Cluster and Computational Grids for Scientific Computing, Faverges, France, Sep. 2002.
- *Deploying and Scheduling Parameter Sweep Applications*, Entropia Inc., San Diego, California, Mar. 2002.
- *From Grid to Global Computing: Deploying and Scheduling Parameter Sweep Applications*, Organization for Parallel Computing Research (ORAP), Saclay, France, Mar. 2002.
- *The Virtual Instrument Project*, National Science Foundation Workshop on Information Technology Research and Education, Palo Alto, California, Mar. 2002.
- *APST and its Uses on Clusters*, IEEE Cluster 2001 Conference, Cluster Application Panel, Newport Beach, California, Oct. 2001.
- *Promoting the Grid with APST*, 1st US/UK Grid Workshop, San Francisco, California, Aug. 2001.
- *Simulation for Application Scheduling*, Monash University, Melbourne, Australia, May 2001.
- *The AppLeS Parameter Sweep Template: User-Level Middleware for the Grid*, NASA IPG workshop, NASA Ames Research Center, Sep. 2000.
- *Internet-wide Scientific Software Access with NetSolve*, **Keynote Address**, Workshop on Internet Accessible Mathematical Computation, Vancouver, Jul. 1999.
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FUNDING AWARDS

- $335,000 / 3 years (PI), National Science Foundation, 08/24 - 07/27
WfCommons Tools for Workflow Research, Development, and Education on the CyberInfrastructure
- $600,000 / 3 years (Co-PI), National Science Foundation, 10/21 - 09/24
Simulation-driven runtime resource management for distributed workflow applications (collaborative award with USC)
- $600,000 / 3 years (Co-PI), National Science Foundation, 08/21 - 07/24
Simulation-driven Evaluation of Cyberinfrastructure Systems (collaborative award with USC)
- $15,000 / 1 years (Co-PI), National Science Foundation, 08/20 - 09/21
Infrastructure for Enabling Systematic Development and Research of Scientific Workflow Management Systems (collaborative award with USC and the University of Chicago)
- $500,000 / 3 years (PI), National Science Foundation, 10/19 - 09/22
Integrating core CI literacy and skills into university curricula via simulation-driven activities (collaborative award with USC)
- $500,000 / 3 years (PI), National Science Foundation, 01/17 - 12/20
WRENCH: A Simulation Workbench for Scientific Worflow Users, Developers, and Researchers (collaborative award with USC)
- $31,764 / 3 years (co-PI), National Science Foundation, 06/09 - 05/12
DiRT: A Testbed for Distributed Research
- $500,000 / 3 years (co-PI), National Science Foundation, 10/05 - 09/08
Designing Large-Scale Distributed Systems for Realistic Failure Models
- $5,400,000 / 5 years (co-PI), National Science Foundation, 10/03 - 09/08
Virtual Grid Application Development Software (VGrADS)
- $500,000 / 3 years (PI), National Science Foundation , 07/03 - 06/06
Models to Support Performance-Engineering of Global Computations
- $48,000 / 3 years (co-PI), National Science Foundation / Institut National de Recherche en Informatique et en Automatique, 07/03 - 06/06
Algorithms and Simulations for Scheduling on Large-Scale Distributed Platforms
- $450,000 / 3 years (co-PI), National Science Foundation, 01/03 - 12/05
Autonomous Scheduling on Large Distributed Systems
- $235,000 / 1 year (PI), National Science Foundation, FY 2004
National Partnership for Advanced Computational Infrastructure (NPACI)
- $750,000 / 3 years (co-PI), National Science Foundation, 09/02 - 08/05
Data Intensive Grid Benchmarks
- $792,000 / 3 years (co-PI), National Science Foundation, 10/02 - 09/05
Optimizing Performance and Reliability in Distributed Computing Systems Through Wide Spectrum Storage Services
- $270,000 / 1 year (PI), National Science Foundation, FY 2003
National Partnership for Advanced Computational Infrastructure (NPACI)

- $700,000 / 2 years (co-PI), National Science Foundation, 07/02 - 06/04
  GridSolve: A System for Grid-enabling General Purpose Problem Solving Environments
- $280,000 / 1 year (PI), National Science Foundation, FY 2002
  National Partnership for Advanced Computational Infrastructure (NPACI)
- $2,500,000 / 3 years (co-PI), National Science Foundation, 09/00 - 08/03
  Virtual Instruments: Scalable Software Instruments for the Grid
- $1,300,000 / 3 years (co-PI), National Science Foundation, 08/99 - 07/02
  Logistical QoS through Application-driven Scheduling of Remote Storage

ADVISING

Ph.D. Dissertation Advising and Co-Advising: Jesse McDonald (2026); Benjamin Karsin (2018); Michael Gowanlock (2015); Mark Stillwell (2010); Joshua Wingstrom (2009); Richard Huang (2007); Derrick Kondo (2005); Yang Yang (2005); Alan Su (2003); Marcio Faerman (2003).

M.S. Thesis Advising and Co-Advising: Will Koch (2021); Evan Hataishi (2019); Lambert Leong (2018); Suraj Pandey (2018); Benjamin Karsin (2012); David Schanzenbach (2011); Kaveh Kardan (2010); Kayo Fujuwara (2007); Krijn van der Raadt (2004); Charles Wurster (2004); Gregory Chun (2003); Otto Sievert (2003); Nadya Williams (2002); Holly Dail (2002); Shava Smallen (2001).

Postdoctoral Researcher Advising: Arnaud Legrand (2005); Gilles Fédak (2004); Frédéric Suter (2003); Yudith Cardinale (2003).

TEACHING

- UHM, ICS312 (UG): Machine-Level and Systems Programming (FA26, SP26, SP25, SP24, FA23, SP23, SP22, FA21, SP21, FA20, SP19, SP18, SP17, SP16, SP15, SP14, SP10, SP09, FA07)
- UHM, ICS432 (UG): Concurrent and High-Performance Programming (FA24, FA22, FA20, SP17, SP16, SP15, FA11, FA08, SP07, SP06)
- UHM, ICS632 (GR): Principles of High Performance Computing (FA21, FA18, FA17, FA16, FA15, FA14, FA08, FA07, FA06, FA05)
- UHM, ICS332 (UG): Operating Systems (SP26, FA23, FA19, SP19, SP18, FA17, SP14, SP13, FA12, SP12, SP11, FA09)
- UHM, ICS690 (GR): Graduate Seminar (SP13, FA12, SP12, FA11, SP11, FA10, SP10, FA09, SP09)
- UHM, ICS431 (UG): Computer Architecture (SP08, FA06)
- UCSD, CSE130 (UG): Programming Languages (SP05, SP04, SP03)
- UCSD, CSE260 (GR): Parallel Computing (FA04)

AWARDS / HONORS

- Chancellor's Citation for Meritorious Teaching, UH Mānoa, 2018.
- Visiting Associate Professor, National Informatics Institute (NII), Japan, 2012-2019.
- Chancellor's Citation for Meritorious Teaching, UH Mānoa, 2012.
- Best paper award at HeteroPar, Aug. 2009.

- Best student paper award at SC, Nov. 2001 (M.S. adviser of Shava Smallen).
- R&D 100 Award for designing and developing the NetSolve project, 1999.

PROFESSIONAL SERVICES

University Services

– Fall 2022 - present: Chair of the Information and Computer Science Undergraduate Committee, University of Hawai'i at Mānoa.

– Fall 2015: Associate Chair of the Information and Computer Sciences Department.

– Spring 2009 - Spring 2013: Chair of the Computer Science Graduate Program, University of Hawai'i at Mānoa.

Conference Chairing

2024

– **Co-Organizing Chair**: 2nd Workshop on Workflows in Distributed Environments, Athens, Greece, April 2024.

2020

– **Algorithm Track Chair**: International Conference on Parallel Processing, Edmonton, Canada, Aug 2020.

2017

– **Local Arrangement Co-Chair**: IEEE Cluster Computing Conference, Honolulu, Hawai'i, Sep. 2017.

2013

– **Co-organizer**: INRIA *Workshop on Scheduling Algorithms for Exascale*, Dagstuhl Schloss, Germany, Sep. 2013.

– **Publicity Chair**: 42nd International Conference on Paralle Processing (ICPP), Lyon, France, Oct. 2013.

2011

– **Program Vice-Chair**: 11th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), Newport Beach, California, May 2011.

2010

– **Program Vice-Chair**: IEEE/ACM Grid Conference, Brussels, Belgium, Oct. 2010.

2009

– **Program Co-Chair**: ACM/IEEE International Symposium on High Performance Distributed Computing (HPDC-18), Garching, Germany, Jun. 2009.

– **Program Vice-Chair**: 9th IEEE Symposium on Cluster Computing and the Grid (CCGrid'09), Shanghai, China, May 2009.

2008

– **Program Co-Chair**: 9th IEEE/ACM International Conference on Grid Computing (Grid 2008), Tokyo, Japan, Nov. 2008.

– **Co-organizer**: NSF/INRIA *Workshop on Scheduling Algorithms for Large-Scale Distributed Platforms*, Aussois, France, May 2008.

– **Publicity Chair**: 8th IEEE Symposium on Cluster Computing and the Grid (CCGrid'08), Lyon, France, May 2008.

– **General Chair**: Heterogeneous Computing Workshop (HCW'08), Miami, Florida, Apr. 2008.

2007

– **Program Topic Chair**: Euro-Par 2007 (Scheduling and Load Balancing topic), Rennes, France, Aug. 2007.

– **Program Chair**: Heterogeneous Computing Workshop (HCW'07), Long Beach, California, Mar. 2007.

2005

– **Program Vice-Chair**: 6th International Workshop on Grid Computing (Grid'05), Seattle, Washington, Nov. 2005.

– **Co-organizer**: NSF/INRIA *Workshop on Scheduling Algorithms for Large-Scale Distributed Platforms*, San Diego, California, Nov. 2005.

2004

– **General Chair**: *IEEE Conference on Cluster Computing (Cluster'04)*, San Diego, Sep. 2004.

– **Co-organizer**: NSF/INRIA *Workshop on Scheduling Algorithms for Large-Scale Distributed Platforms*, Aussois, France, Aug. 2004.

2003

– **Invited Speaker Chair**: International Conference for High Performance Computing and Communications (SC'03), Phoenix, Arizona, Nov. 2003.

1999

– **Organizer**: Mini-symposium on Network-enabled Servers, SIAM Annual Meeting 1999, Atlanta, Georgia, May 1999.

Journals

– 2012-2016: **Subject Area Editor** for the *Parallel Computing* journal.

– 2004-2009: Member of the **Editorial Board** of the *Future Generation Computer Systems* journal.

– 2006: **Guest Editor** for a special issue of the *International Journal of High Performance Computing and Applications*.

– 2005: **Guest Editor** for the a special issue of the *IEEE Transactions on Parallel and Distributed Systems (TPDS)*.

Conference Program Committees

2026

– 21st Workshop on Workflows in Support of Large-Scale Science (WORKS), Chicago, Illinois, Nov 2026.

– 29th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), New Orleans,

U.S.A., May 2026.

- International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), Panel Committee, Nov 2026.

2025

- 20th Workshop on Workflows in Support of Large-Scale Science (WORKS), St. Louis, Missouri, Nov 2025
- 33rd Euromicro/IEEE International Conference on Parallel, Distributed, and Network-Based Processing, Turin, Italy, March 2025.
- 28th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Milan, Italy, May 2025.

2024

- 8th International Workshop on Software Correctness for HPC Applications, Atlanta, Georgia, November 2024
- 19th Workflows in Support of Large-Scale Science (WORKS) Workshop, Atlanta, Georgia, November 2024.
- International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), Nov 2024
- International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), Nov 2024.
- 27th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), San Francisco, California, May 2024.

2023

- 37th IEEE International Parallel and Distributed Processing Symposium (IPDPS), St. Petersburg, Florida, May 2023
- 26th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), St. Petersburg, Florida, May 2022.

2022

- International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), Dallas, Texas, Nov. 2022.
- 17th Workflows in Support of Large-Scale Science (WORKS) Workshop, Dallas, Texas, November 2022.
- 36th IEEE International Parallel and Distributed Processing Symposium (IPDPS), Lyon, France, May 2022.
- 25th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Lyon, France, June 2022.

2021

- 16th Workflows in Support of Large-Scale Science (WORKS) Workshop, Saint Louis, Missouri, November 2021.
- 27th International European Conference on Parallel and Distributed Computing (Europar), Lisbon, Portugal, August 2021.

– 50th International Conference on Parallel Processing (ICPP), Chicago, Illinois, August 2021.

– Conférence francophone d'informatique en Parallélisme, Architecture et Systèmes, Lyon, France, July 2021.

– 35th IEEE International Parallel and Distributed Processing Symposium (IPDPS), Porland, Oregon, May 2021.

2020

– 15th Workflows in Support of Large-Scale Science (WORKS) Workshop, Georgia, Atlanta, November 2020.

– Conférence francophone d'informatique en Parallélisme, Architecture et Système (COMPAS), Lyon, France, June 2020.

– 34rd IEEE International Parallel and Distributed Processing Symposium (IPDPS), New Orleans, Louisiana, May 2020.

2019

– 14th Workflows in Support of Large-Scale Science (WORKS) Workshop, Denver, Colorado, November 2019.

– 23rd Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Rio de Janeiro, Brazil, May 2019.

– 19th IEEE International Symposium on Cluster, Cloud and Grid Computing (CCGrid), Larnaca, Cyprus, May 2019.

– 33rd IEEE International Parallel and Distributed Processing Symposium (IPDPS), Rio de Janeiro, Brazil, May 2019.

2018

– 13th Workflows in Support of Large-Scale Science (WORKS) Workshop, Dallas, Texas, November 2018.

– 22nd Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Vancouver, Canada, May 2018.

– 25th IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC), Bengaluru, India, December 2018.

2017

– 12th Workflows in Support of Large-Scale Science (WORKS) Workshop, Denver, Colorado, November 2017.

– 24th IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC), Jaipur, India, December 2017.

– 21st Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Orlando, Florida, June 2017.

– 31st IEEE International Parallel and Distributed Processing Symposium (IPDPS), Orlando, Florida, May 2017.

– 17th IEEE International Symposium on Cluster, Cloud and Grid Computing (CCGrid), Madrid, Spain, May 2017.

2016

– 20th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Hyderabad, Chicago, Illinois, May 2016.

– 1st Workshop on Fault Tolerance for Extreme Scale Computing, Kyoto, Japan, May 2016.

– 30th IEEE International Parallel and Distributed Processing Symposium (IPDPS), Chicago, Illinois, May 2016.

– 45th International Conference on Parallel Processing (ICPP), Philadephia, Pennsylvania, Aug. 2016.

2015

– Workshop on Scheduling for Parallel Computing, Krakow, Poland, Sep. 2015.

– EuroPar Conference, Scheduling and Load Balancing track, Vienna, Austria, Aug. 2015.

– Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Hyderabad, India, May 2015.

– 29th IEEE International Parallel and Distributed Processing Symposium (IPDPS), Hyderabad, India, May 2015.

– 44th International Conference on Parallel Processing (ICPP), Paris, France, Mar. 2015.

– 9th ACM Richard Tapia Celebration of Diversity in Computing (Tapia2015), Boston, Massachusetts, Feb. 2015.

2014

– Workshop on Reproducible Parallel Computing, Porto, Portugal, Aug. 2014.

– Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Phoenix, Arizona, May 2014.

– International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), New Orleans, Louisiana, Nov. 2014.

2013

– 42nd International Conference on Parallel Processing (ICPP), Lyon, France, Oct. 2013.

– 1st conference on Energy Efficiency on Large Scale Distributed Systems (EE-LSDS), Vienna, Austria, Apr. 2013.

– 17th Workshop on Job Scheduling Strategies for Parallel Processing, Boston, Massachusetts, May 2013.

– Workshop on Scheduling for Parallel Computing, Warsaw, Poland, Sep. 2013.

– 13th IEEE International Symposium on Cluster, Cloud and Grid Computing (CCGrid), Delft, The Netherlands, May 2013.

– IEEE Computer Society's Technical Committee on Parallel Processing's Ph.D. Forum at IPDPS, Boston, Massachusetts, May 2013.

– 27th International Parallel and Distributed Processing Symposium (IPDPS), Boston, Massachusetts , May 2013.

– 22nd IEEE Symposium on High-Performance Distributed Computing (HPDC-22), New York, New York, Jun. 2013.

2012

– IEEE Cluster Conference, Beijing, China, Sep. 2012.

– 3rd Workshop on Scientific Cloud Computing, Delft, The Netherlands, Jun. 2012.
– 41st International Conference on Parallel Processing (ICPP), Pittsburgh, Pensylvania, Sep. 2012.
– 25th IEEE International Symposium on Computer-Based Medical Systems (CBMS), Rome, Italy, Jun. 2012.
– 26th International Parallel and Distributed Processing Symposium (IPDPS), Shanghai, China, May 2012.
– IEEE Computer Society's Technical Committee on Parallel Processing's Ph.D. Forum at IPDPS, Shanghai, China, May 2012.
– 12th IEEE International Symposium on Cluster, Cloud and Grid Computing (CCGrid), Ottawa, Canada, May 2012.
– 16th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Shanghai, China, May 2012.

2011

– IEEE Cluster Conference, Austin, Texas, Sep. 2011.
– Workshop on Scheduling for Parallel Computing, Torun, Poland, Sep. 2011.
– 40th International Conference on Parallel Processing (ICPP), Taipei, Taiwan, Sep. 2011.
– 2nd Workshop on Scientific Cloud Computing (ScienceCloud), San Jose, California, Jun. 2011.
– IEEE Computer Society's Technical Committee on Parallel Processing's Ph.D. Forum at IPDPS, Anchorage, Alaska, May 2011.
– 5th Workshop on Desktop Grids and Volunteer Computing Systems (PCGrid), Anchorage, Alaska, May 2011.
– 25th International Parallel and Distributed Processing Symposium (IPDPS), Anchorage, Alaska, May 2011.

2010

– 17th IEEE International Conference on High Performance Computing (HiPC), Goa, India, Dec. 2010.
– International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), New Orleans, Louisiana, Nov. 2010.
– 3rd Workshop on Many-Task Computing on Grids and Supercomputers, (MTAGS), New Orleans, Louisiana, Nov. 2010.
– 4th International Workshop on HealthGrid & Cloud Computing, Perth, Australia, Oct. 2010.
– 1st Workshop on Scientific Cloud Computing (ScienceCloud), Chicago, Illinois, Jun. 2010.
– 19th IEEE Symposium on High-Performance Distributed Computing (HPDC-19), Chicago, Illinois, Jun. 2010.
– IEEE Computer Society's Technical Committee on Parallel Processing's Ph.D. Forum at IPDPS, Atlanta, Georgia, May 2010.
– 4th Workshop on Desktop Grids and Volunteer Computing Systems (PCGrid), Melbourne, Australia, May 2010.
– 10th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid),

Melbourne, Australia, May 2010.

- 15th Workshop on Job Scheduling Strategies for Parallel Processing (JSSPP), Atlanta, Georgia, Apr. 2010.

2009

- International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), Portland, Oregon, Nov. 2009.
- Workshop on Scheduling for Parallel Computing, Krakow, Poland, Sep. 2009.
- 8th International Symposium on Parallel and Distributed Computing (ISPDC), Lisbon, Portugal, Jun. 2009.
- Workshop on Grids, Clouds and Virtualization (WGCV), Geneva, Switzerland, May 2009.
- Fourth Workshop on System Management Tools for Large-Scale Parallel Systems (SMTPS), Rome, Italy, May 2009.
- 8th IEEE International Symposium on Cluster Computing and the Grid (CCGrid), Shanghai, China, May 2009.
- 3rd Workshop on Desktop Grids and Volunteer Computing Systems (PCGrid), Rome, Italy, May 2009.
- IEEE Computer Society's Technical Committee on Parallel Processing's Ph.D. Forum at IPDPS, Rome, Italy, May 2009.
- 2009 ACM Symposium on Applied Computing (SAC), Honolulu, Hawaii, U.S.A., Mar. 2009.

2008

- High Performance Data Grid (HPDataGrid'08) workshop, Dunedin, New Zealand, Dec. 2008.
- 1st International Workshop on Robust Scheduling (RobShed'08), Melbourne, Australia, Dec. 2008.
- 14th IEEE International Conference on Parallel and Distributed Systems (ICPADS'08), Melbourne, Australia, Dec. 2008.
- International Symposium on Parallel and Distributed Processing and Applications (ISPA'08), Sydney, Australia, Sep. 2008.
- 17th IEEE Symposium on High-Performance Distributed Computing (HPDC-17), Boston, Massachusetts, Jun. 2008.
- 1st Workshop on Assessing Models of Networks and Distributed Computing Platforms, Lyon, France, May 2008.
- 8th IEEE Symposium on Cluster Computing and the Grid (CCGrid'08), Lyon, France, May 2008.
- International Symposium on Parallel Algorithms, Architectures, and Networks (ISPAN), Sidney, Australia, May 2008.
- Third Workshop on System Management Tools for Large-Scale Parallel Systems (SMTPS), Miami, Florida, Apr. 2008.
- 41st Hawai'i International Conference on System Sciences (HICSS), Software Technology Mini-Track on Grid Computing: Adaptive Software, Tools and Applications, O'ahu, Hawaii, Jan. 2008.

2007

- International Conference on Parallel Processing (ICPP), Xi'an, China, Sep. 2007.
- 1st Workshop on Scheduling for Parallel Computing, Gdansk, Poland, Sep. 2007.
- 16th International Conference on Computer and Networks (ICCCN), O'ahu, Hawai'i, Aug. 2007.
- International Conference on Distributed Computing Systems (ICDCS), Toronto, Canada, Jun. 2007.
- 16th IEEE Symposium on High Performance and Distributed Computing (HPDC-16), Monterey, California, Jun. 2007.
- 7th workshop on Global and Peer-to-Peer Computing (GP2PC), Rio de Janeiro, Brazil, May 2007.
- Proposal Reviewer and Panel Participant, National Science Foundation, Washington D.C., Apr. 2007.
- Third Workshop on System Management Tools for Large-Scale Parallel Systems (SMTPS), Long Beach, California, Mar. 2007.

2006

- 12th IEEE International Conference on Parallel and Distributed Systems (ICPADS'06), Minneapolis, Minnesota, Jul. 2006.
- 15th IEEE Symposium on High Performance and Distributed Computing (HPDC-15), Paris, France, Jun. 2006.
- Proposal Reviewer and Panel Participant, National Science Foundation, Washington D.C., Apr. 2006.
- Third High Performance Grid Computing Workshop (HPGCW'06), Rhodes, Greece, Apr. 2006.
- Heterogeneous Computing Workshop (HCW'06), Rhodes, Greece, Apr. 2006.
- Second Workshop on System Management Tools for Large-Scale Parallel Systems (SMTPS), Rhodes, Greece, Apr. 2006.
- 5th Workshop on Global and Peer-to-Peer Computing (WGP2PC'06), Singapore, Apr. 2006.

2005

- International Conference for High Performance Computing and Communications (SC'05), Seattle, Washington, Nov. 2005.
- Program Committee for the 1st International Workshop on Parallel Bioinspired Algorithms, Oslo, Norway, Jun. 2005
- IEEE International Conference on Pervasive Services (ICPS 2005), Santorini, Greece, Jun. 2005.
- 4th Workshop on Global and Peer-to-Peer Computing (WGP2P'05), Cardiff, U.K., May 2005.
- First Workshop on System Management Tools for Large-Scale Parallel Systems, Denver, Colorado, Apr. 2005.
- Heterogeneous Computing Workshop (HCW'05), Denver, Colorado, Apr. 2005.
- High Performance Grid Computing Workshop, Denver, Colorado, Apr. 2005.

2004

- International Conference for High Performance Computing and Communications (SC'04), Pittsburgh, Pennsylvania, Nov. 2004.
- 17th International Conference on Parallel and Distributed Computing Systems (PDCS 2004), San Francisco, California, Sep. 2004.
- 13th IEEE Symposium on High Performance and Distributed Computing (HPDC-13), Honolulu, Hawai'i, Jun. 2003.
- 3rd Workshop on Global and Peer-to-Peer Computing (WGP2P'04), Chicago, Illinois, Apr. 2004.

2003

- International Conference for High Performance Computing and Communications (SC'03), Phoenix, Arizona, Nov. 2003.
- Proposal Reviewer and Panelist, National Science Foundation, Washington, DC., Oct. 2003.
- Program Committee member for Euro-Par 2003, Klagenfurt, Austria, Aug. 2003.
- Workshop on Innovative Solutions for Grid Computing (InnoGrid'03), Melbourne, Australia, Jun. 2003.
- 12th IEEE Symposium on High Performance and Distributed Computing (HPDC-12), Seattle, Washington, Jun. 2003.
- ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP), San Diego, California, Jun. 2003.
- 3rd IEEE Symposium on Cluster Computing and the Grid (CCGrid'03), Tokyo, Japan, May 2003.
- 3rd Workshop on Global and Peer-to-Peer Computing (WGP2P'03), Tokyo, Japan, May 2003.
- International Parallel and Distributed Processing Symposium (IPDPS'03), Nice, France, Apr. 2003.

2002

- 3rd International Workshop on Grid Computing (Grid'02), Baltimore, Maryland, Nov. 2002.
- International Conference for High Performance Computing and Communications (SC'02), Baltimore, Maryland, Nov. 2002.
- 11th IEEE Symposium on High Performance and Distributed Computing (HPDC-11), Edinburgh, U.K., Aug. 2002.
- 2nd Workshop on Global and Peer-to-Peer Computing (WGP2P'02), Berlin, Germany, May 2002.

2001

- 2nd International Workshop on Grid Computing (Grid'01), Denver, Colorado, Nov., 2001.
- Metacomputing Systems and Applications Workshop (MSA'01), Valencia, Spain, Sep. 2001.
- Euro-Par'01 Conference, Manchester, U.K., Aug. 2001.
- High Performance Computing (HPC'01) Conference, Seattle, Washington, Apr. 2001.